Arkaprava Gupta

Chapel Hill, NC | arka@cs.unc.edu | 919-685-2804
LinkedIn: linkedin.com/in/arkaprava-gupta
Google Scholar: scholar.google.com/citations?user=28KcAy8AAAAJ

Research Interests

- Cyber-Physical Systems and Autonomous Control
- Resource-Constrained and Embedded Machine Learning
- Formal Methods for Safety-Critical Systems

Education

University of North Carolina at Chapel Hill
Ph.D. in Computer Science | Aug 2024 - Present

- GPA: **4.0**/4.0

Kalinga Institute of Industrial Technology (KIIT) University, India
Bachelor of Technology (Honours) in Computer Science | Aug 2020 - Jun 2024

- GPA: **9.34**/10.00

Experience

Research Assistant
The University of North Carolina at Chapel Hill | Chapel Hill, NC
Aug 2024 - Present

Supervisors: Dr. Sridhar Duggirala and Prof. Samarjit Chakraborty
Skills: Python, Julia, MATLAB, CARLA, Scenic, PyTorch, TensorFlow, Language Models, Reinforcement Learning, Multi-Armed Bandit, Controller Design, Microcontroller Programming (C/C++), GitHub.

- Designed test suites by reconstructing autonomous driving scenarios in simulated environments from videos in the NuScenes dataset to facilitate systematic evaluation of perception and control pipelines.
- Developed and deployed on hardware resource-constrained RL algorithms for closed-loop Deep Brain Stimulation (DBS) under real-time and energy constraints to be used within an implant.
- Designed a hybrid edge-cloud control strategy that combines low-latency edge inference with high-accuracy cloud inference that outperforms standalone edge-only and cloud-only controllers in autonomous vehicles.

Research Intern
Ericsson India Pvt. Ltd. | Bengaluru, India
Sep 2023 - Jun 2024

Supervisor: Dr. Swarup Mohalik
Skills: Python, Spiking Neural Networks, Formal Methods, snnTorch, Nengo, Gurobi, CPLEX, GitHub.

- Trained Spiking Neural Networks (SNNs) for various ML problems; SNNs are an extremely power-efficient alternative to traditional neural networks.
- Proposed a Mixed-Integer Linear Programming (MILP) encoding to formally verify SNNs with the Spiking Rectified Linear Activation (SRLA) neurons.
- Designed an algorithm to use these encodings to synthesize safe SNN controller configurations as replacements for given controllers that used traditional deep neural networks.

Summer Research Fellow
Indian Statistical Institute | Kolkata, India
May 2023 - Jul 2023

Supervisors: Prof. Ansuman Banerjee and Dr. Sumana Ghosh
Skills: Python, Spiking Neural Networks (SNNs), Formal Methods, Z3, snnTorch, PyTorch.

- Performed output range analysis of SNNs to assess safety for deployment in cyber-physical systems using formal verification.
- Developed a pruning technique for the deployment of SNNs on hardware in an attempt to reduce space on hardware, increase speed, enable more scalable verification and lower energy consumption.

Publications and Research

1. **Controller-aware Perception Neural Network Pruning for CPS Safety Optimization**
   J. Chen, Y. Song, S. Xu, K. Kessler, P. Ganguli, **A. Gupta**, B. Ghosh, P. S. Duggirala, T. Chen, S. Chakraborty, J. Hu.
   ACM Transactions on Cyber-Physical Systems (TCPS), 2026.

2. **Bandit Algorithms for Deep Brain Stimulation**
   A. Gupta, N. Carter, W. Zellers, P. Ganguli, B. Dietrich, V. Krishna, P. S. Duggirala, S. Chakraborty.
   Proceedings of the 17th ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS 2026), Saint-Malo, France, May 11-14, 2026.

3. **Building Safe Autonomous Systems Using Imperfect Components**
   S. Xu, P. Ganguli, T. Zhu, **A. Gupta**, B. Ghosh, K. Wilson, A. Al Arafat, J. Baugh, Z. Guo, B. Berg, P. S. Duggirala, S. Chakraborty.
   Engineering Trustworthy Software Systems (SETSS 2025), Lecture Notes in Computer Science, vol. 16481, Springer, pp. 213-286, 2026.

4. **Navigating Delay-Uncertainty Tradeoffs: Controller Design for DNN-Based Edge-Cloud Perception**
   P. Ganguli, T. Zhu, L. Wang, **A. Gupta**, S. Xu, P. S. Duggirala, S. Chakraborty.
   33rd International Conference on Real-Time Networks and Systems (RTNS 2025), Pisa, Italy, November 5-7, 2025.

5. **In-Vivo Training for Deep Brain Stimulation**
   N. Carter, **A. Gupta**, P. Ganguli, B. Dietrich, V. Krishna, S. Chakraborty.
   Proceedings of the IEEE 21st International Conference on Body Sensor Networks (BSN 2025), Track: Optimization and Personalization for Computational Health, November 3-5, 2025.

6. **Controllers for Edge-Cloud Cyber-Physical Systems**
   T. Zhu, P. Ganguli, **A. Gupta**, S. Xu, L. Capogrosso, E. Fraccaroli, M. Cristani, F. Fummi, S. Chakraborty.
   Proceedings of the 17th International Conference on Communication Systems and Networks (COMSNETS 2025), Bengaluru, India, January 6-10, 2025.

7. **Configuring Safe Spiking Neural Controllers for Cyber-Physical Systems through Formal Verification**
   A. Gupta, S. Ghosh, A. Banerjee, S. K. Mohalik.
   Proceedings of the 22nd ACM/IEEE International Symposium on Formal Methods and Models for System Design (MEMOCODE 2024), part of ESWEEK, Raleigh, NC, USA, October 3-4, 2024.

8. **A Test Generation Approach for Spiking Neural Network Simplification**
   S. Banerjee, **A. Gupta**, S. Ghosh, A. Banerjee, S. K. Mohalik.
   Proceedings of the 27th International Symposium on VLSI Design and Test (VDAT 2023), BITS Pilani, India, September 29-October 1, 2023.

9. **Safety Analysis of SNN-controlled Cyber-Physical Systems**
   A. Gupta, S. Ghosh, A. Banerjee.
   Cyber-Physical Systems Summit (CyPhySS 2023), 7th Annual Symposium on Cyber-Physical Systems, IIT Kharagpur, India, July 20-22, 2023.

Projects

1. **Domain Randomization for Zero-Shot Sim-to-Real Transfer in Robotics**
   Skills: Python, Reinforcement Learning, Robotics Simulation, Domain Randomization, Sim-to-Real Transfer.
   Implemented and evaluated domain randomization (DR) strategies for *sim-to-real* transfer in robotics from PyBullet to MuJoCo. Identified key parameters crucial for DR through systematic parameter ablations.
2. **Universal vs. Specialized LoRA Adapters for LLM Transfer Learning**
   Skills: Python, LLMs, Parameter-Efficient Fine-Tuning (PEFT), Multi-Task Learning, Transfer Learning.
   Compared a transferred single multi-task LoRA adapter against transferred task-specific adapters with routing across GSM8K, LeetCode, CNN/DailyMail, and SimpleQA, showing that a universal adapter trained on <1% parameters remains competitive while specialized adapters yield peak task-specific gains.
3. **Biomedical Image Classification for Lung Cancer**
   Skills: Python, Computer Vision, Deep Learning, Medical Image Analysis, CNN Architectures.
   Evaluated and adapted standard deep learning architectures for medical image classification, experimenting with architectural modifications to improve lung cancer detection accuracy.

Technical Skills

Languages: Python, MATLAB, R, Julia, Java, C++, C
ML & Data Analysis: PyTorch, TensorFlow, Keras, snnTorch, Nengo, NumPy, Pandas, Matplotlib
Formal Methods & Optimization: Z3, Gurobi, CPLEX
Systems & Tools: AWS, Docker, Android Studio